CM CASE: LCI MULTIPLICITY EQUALS THE LENGTH OF THE SCHEME-THEORETIC INTERSECTION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file intersection.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 0 hits for this chapter (is-occupied.json, 2026-10-14); consistent with the duty manager's six-check dispatch.

Source reference. Original lemma 0B04: sources/stacks-chow/tags.txt, source lines 13847–13847. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0B02: sources/stacks-chow/tags.txt, source lines 13845–13845. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 02JN: sources/stacks-chow/tags.txt, source lines 3342–3342. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00LN: sources/stacks-chow/tags.txt, source lines 793–793. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *In Lemma 0B04 assume that the local ring $\mathcal{O}_{W,\xi}$ is Cohen-Macaulay. Then we have*

$$e(X, V \cdot W, Z) = \mathit{length}_{\mathcal{O}_{X,\xi}}(\mathcal{O}_{W,\xi}/f_1\mathcal{O}_{W,\xi} + \ldots + f_c\mathcal{O}_{W,\xi}).$$

Proof. This follows immediately from Lemma 0B02. Alternatively, we can deduce it from Lemma 0B04. Namely, by Algebra, Lemma 02JN we see that $f_1, \ldots, f_c$ is a regular sequence in $\mathcal{O}_{W,\xi}$. Then Algebra, Lemma 00LN shows that $f_1, \ldots, f_c$ is a quasi-regular sequence. This easily implies the length of $\mathcal{O}_{W,\xi}/(f_1, \ldots, f_c)^t$ is

$$\binom{c+t}{c}\text{length}_{\mathcal{O}_{X,\xi}}(\mathcal{O}_{W,\xi}/f_1\mathcal{O}_{W,\xi} + \ldots + f_c\mathcal{O}_{W,\xi}).$$

Looking at the leading coefficient we conclude. □

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